AProjectReport On

Sports performance analysis

Submittedinpartialfulfillmentoftherequirementforthe award of the degree

MASTEROFSCIENCE(DATASCIENCE)
from
MarwadiUniversity

AcademicYear2026–27

Nevil Aghera (92500584012)
Jenil Kothiya (92500584026)

InternalGuide
Gehna Sachdeva

Rajkot-MorbiRoad,At&PO:Gauridad,Rajkot360003.Gujarat.India.

FacultyofComputerApplications(FoCA)

Certificate

This is to certify that the project work entitled

HeartFailurePredictionUsingLogisticRegression

submittedin partialfulfillmentoftherequirementfor theawardofthedegreeof

MasterofScience(DataScience)

of the

MarwadiUniversity

isaresultofthebonafideworkcarriedout by

Nevil Aghera (92500584012)

Jenil Kothiya (92500584026)

duringtheacademicyear2025–2026

Gehna Sachdeva	*Dr.SunilBajeja*	*Dr.Rizvan Khan*
FacultyGuide	**HOD**	**Dean**

DECLARATION

We hereby declare that this project work entitled **Cricket Player Performance**is a record done by us.

Wealsodeclarethatthematterembodiedinthisprojectisgenuineworkdoneby us and has not been submitted whether to this University or to any other University/Instituteforthefulfillmentoftherequirementofanycourseofstudy.

Place:Rajkot

Date:

Nevil Aghera (92500584012) Signature:_____
Jenil Kothiya(92500584026) Signature:_____

ACKNOWLEDGEMENT

It is indeed a great pleasure to express our thanks and gratitude to all those who helped us. No serious and lasting achievement or success one can ever achieve without the help of friendly guidance and co-operation of so many people involved in the work.

Weareverythankfultoourguide**Gehna Sachdeva,**thepersonwho makes us to follow the right steps during our project work. We express our deep sense of gratitude to for his guidance, suggestions and expertise at every stage. A part from that his valuable and expertise suggestion during documentation of our report indeed help us a lot.

Thanks to our friend and colleague who have been a source of inspiration and motivation that helped to us during our project work.

We are heartily thankful to the Dean of our department **Dr. Rizvan Khan**for givingusanopportunitytoworkoverthisprojectandfortheirend-lessandgreat support to all other people who directly or indirectly supported and help us to fulfil our task.

Nevil Aghera (92500584012) Signature:_____
Jenil Kothiya(92500584026) Signature:_____

CONTENTS

1. Introduction:

1.1 Objective of the New System

- The objective of this system is to analyze cricket player performance data and generate meaningful insights such as batting form, strike rate trends, and impact scores, helping coaches, analysts, and selectors make data-driven decisions.

1.2 Problem Definition

- Currently, player performance evaluation is done manually using scattered scorecards, which is time-consuming and prone to inconsistency. There is a need for a centralized system that automatically processes match data and provides accurate, quick performance analysis.

1.3 Core Components

- The system consists of a data collection module (match records), a processing/analysis engine (statistics, impact score calculation), a database to store player and match data, and a reporting/visualization module to present results.

1.4 Project Profile

- The project is a data-driven Cricket Player Performance Analysis System built to process historical match data (runs, wickets, strike rate, economy rate, etc.) and generate performance summaries and predictive insights for players across match formats (T20, ODI, Test).

1.5 Assumptions and Constraints

- It is assumed that input data is accurate and complete. The system is constrained by the availability of historical data and does not account for real-time weather or pitch conditions during analysis.

1.6 Advantages and Limitations of the Proposed System

- **Advantages***:* Fast and accurate analysis, reduces manual effort, supports better decision-making.

- **Limitations***:* Depends on data quality; does not factor in psychological or situational match pressure on players.

2. RequirementDetermination&Analysis:

2.1Requirement Determination

- Historical match data required (runs, balls faced, fours, sixes, wickets, economy rate, strike rate, match result).
- Data cleaning and preprocessing to handle missing/incorrect values.
- Exploratory Data Analysis (EDA) to understand patterns and trends.
- Feature engineering to create relevant input variables for the model.
- Build ML model to predict player performance/impact score.
- Visualize results for easy interpretation by non-technical users.

2.2Targeted Users

- Cricket team analysts
- Coaches and selectors
- Sports journalists
- Fantasy league players
- Sports management organizations

2.3Tool Details (Python / Power BI / Tableau)

- Python: Data preprocessing, EDA, and ML model building/training.
- Power BI / Tableau: Interactive dashboards for visualizing player stats and trends.
- Jupyter Notebook: Development environment for coding and testing.

2.4Library Description

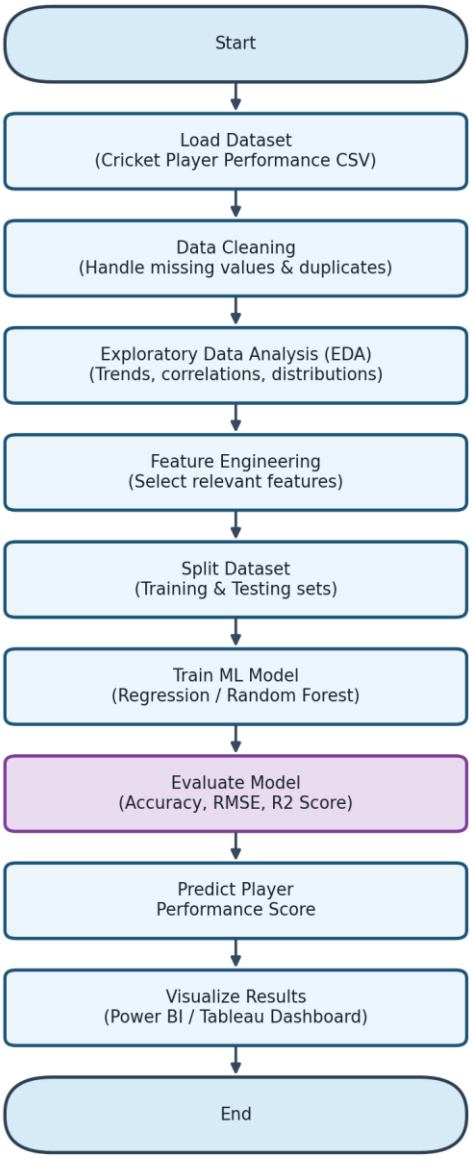
- Pandas: Data loading, cleaning, and manipulation.
- NumPy: Numerical computations on dataset.
- Matplotlib / Seaborn: Data visualization (graphs, charts, correlations).
- Scikit-learn: ML model building (Regression/Random Forest) and evaluation.
- Jupyter Notebook: Interactive coding and step-by-step analysis

3. SystemDesign:

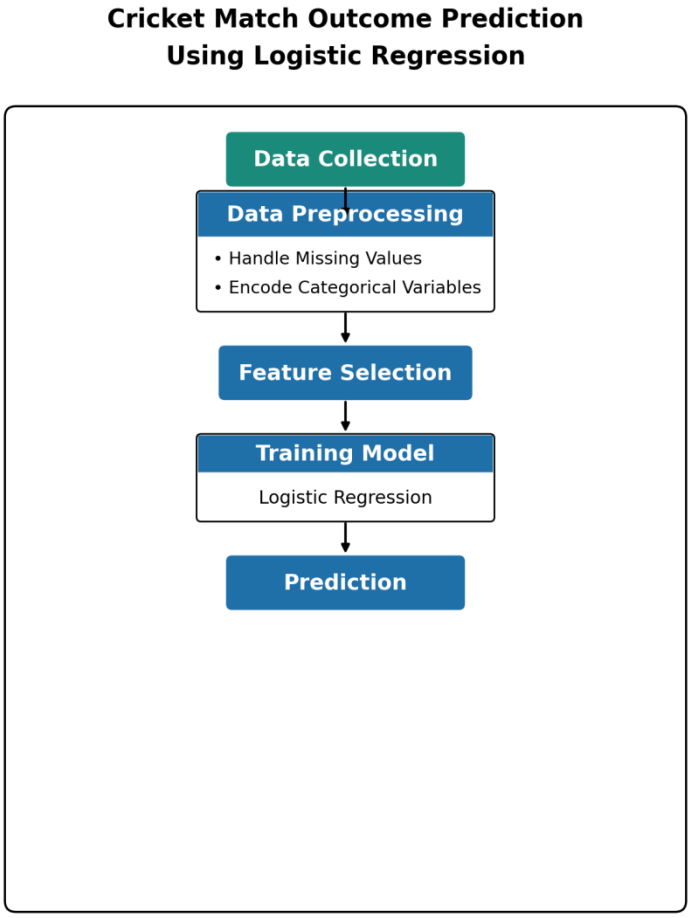
3.1Flowchart/Algorithmwith steps:

- Step 1: Load dataset
- Step 2: Replace missing values using SimpleImputer
- Step 3: Encode categorical values using LabelEncoder
- Step 4: Scale numerical values using MinMaxScaler
- Step 5: Split data into training and test sets
- Step 6: Train models using different algorithms
- Step 7: Evaluate model performance using metrics
- Step 8: Visualize data using heatmap, scatter, boxplot, and histogram

➢ **Flowchart:**



3.2DatasetDesign:



The dataset includes 40,000 real-world style samples of individual player performances across international cricket matches, with attributes such as runs scored, balls faced, boundaries (fours and sixes), wickets taken, economy rate, strike rate, catches, and target score. The target variable is "Match_Result", which classifies the outcome of the match as Win, Loss, or Tie.

3.3Details on preprocessing steps applied:

- **Missing Values:** Dataset checked for nulls; none were found. As a precaution, SimpleImputer (mean for numerical, most frequent for categorical) is applied in the pipeline.
- **Categorical Encoding:** LabelEncoder applied to all object type columns (Player_Name, Team, Opponent, Match_Type, Match_Result).
- **Feature Scaling:** MinMaxScaler used on numerical columns (Runs, Balls_Faced, Fours, Sixes, Wickets, Economy_Rate, Player_Impact_Score, Strike_Rate, Catches, Target_Score) to normalize them between 0 and 1.
- **Date Handling:** The Date column is parsed and decomposed into Year, Month, and Day features before dropping the original column.

4. Development:

4.1Source Code / Output

- **Source Code :**

 # Cricket Player Performance

1. Import Tools
"""

```
# Commented out IPython magic to ensure Python compatibility.
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder, MinMaxScaler
from sklearn.impute import SimpleImputer

from sklearn.svm import SVC
from sklearn.naive_bayes import GaussianNB
from sklearn.neighbors import KNeighborsClassifier
from sklearn.tree import DecisionTreeClassifier
from sklearn.ensemble import RandomForestClassifier

from sklearn.metrics import confusion_matrix, accuracy_score, classification_report

# %matplotlib inline
pd.set_option('display.max_columns', None)
```

"""## 2. Import Data"""

```
df = pd.read_csv('cricket_player_ performance.csv')
df
```

"""## 3. HBIS — Head, Basic Info & Shape"""

```
# Head - first 5 rows
df.head()

# Basic Info - column dtypes & non-null counts
df.info()

# Shape - rows x columns
print("Shape:", df.shape)

# Statistical Summary
df.describe()
```

"""## 4. Train Data — Clean, EDA, Label Encode, Split

4.1 Data Cleaning (Missing Values & Duplicates)

```
# Replace placeholder characters with NaN
df.replace('?', np.nan, inplace=True)

# Check missing values
print("Missing values per column:\n", df.isnull().sum())
print("\nTotal missing values:", df.isnull().sum().sum())

# Check duplicates
print("\nDuplicate rows:", df.duplicated().sum())
df.drop_duplicates(inplace=True)

# Separate numeric and categorical columns
numerical_cols = df.select_dtypes(include=['number']).columns
categorical_cols = df.select_dtypes(include=['object', 'string']).columns

# Impute (safety step, applies only if missing values exist)
if df[numerical_cols].isnull().sum().sum() > 0:
df[numerical_cols] = SimpleImputer(strategy='mean').fit_transform(df[numerical_cols])

if df[categorical_cols].isnull().sum().sum() > 0:
df[categorical_cols] = SimpleImputer(strategy='most_frequent').fit_transform(df[categorical_cols])

print("Remaining missing values:", df.isnull().sum().sum())
```

"""### 4.2 Feature Engineering — Extract Date Parts"""

```
df['Date'] = pd.to_datetime(df['Date'])
df['Match_Year'] = df['Date'].dt.year
df['Match_Month'] = df['Date'].dt.month
df.drop(columns=['Date'], inplace=True)
df.head()
```

"""### 4.3 Exploratory Data Analysis (EDA)"""

```
# Target class distribution
print(df['Match_Result'].value_counts())

plt.figure(figsize=(6, 4))
sns.countplot(data=df, x='Match_Result')
plt.title("Match Result Distribution")
plt.show()

# Match type distribution
plt.figure(figsize=(6, 4))
sns.countplot(data=df, x='Match_Type')
plt.title("Match Type Distribution")
plt.show()
```

```
# Correlation among numeric features
plt.figure(figsize=(6, 5))
sns.heatmap(df.select_dtypes(include=['number']).corr(), annot=True, cmap="coolwarm", fmt=".2f", linewidths=0.5)
plt.title("Feature Correlation Heatmap")
plt.show()

"""### 4.4 Label Encoding (Categorical Columns)"""

categorical_cols = df.select_dtypes(include=['object', 'string']).columns
label_encoders = {}
for col in categorical_cols:
le = LabelEncoder()
df[col] = le.fit_transform(df[col])
label_encoders[col] = le

print("Match_Result classes:", dict(enumerate(label_encoders['Match_Result'].classes_)))
df.head()

"""### 4.5 Feature Scaling"""

target = 'Match_Result'
numerical_cols = df.select_dtypes(include=['number']).columns
scale_cols = [c for c in numerical_cols if c != target]

scaler = MinMaxScaler()
df[scale_cols] = scaler.fit_transform(df[scale_cols])
df.head()

"""### 4.6 Train-Test Split"""

features = df.columns[df.columns != target]
X = df[features]
y = df[target]

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.3, random_state=0)

print("Training set:", X_train.shape)
print("Testing set:", X_test.shape)

"""## 5. Fit Algorithms on Train Dataset & Evaluate One by One

### 5.1 SVM (Support Vector Machine)
"""

svm_model = SVC(kernel='rbf', C=1, gamma='scale')
svm_model.fit(X_train, y_train)
y_pred_svm = svm_model.predict(X_test)

print("Confusion Matrix (SVM):\n", confusion_matrix(y_test, y_pred_svm))
print("\nAccuracy (SVM):", round(accuracy_score(y_test, y_pred_svm) * 100, 2), "%")
print("\nClassification Report (SVM):\n", classification_report(y_test, y_pred_svm))
```

"""### 5.2 Naive Bayes (NB)"""

```
nb_model = GaussianNB()
nb_model.fit(X_train, y_train)
y_pred_nb = nb_model.predict(X_test)

print("Confusion Matrix (Naive Bayes):\n", confusion_matrix(y_test, y_pred_nb))
print("\nAccuracy (Naive Bayes):", round(accuracy_score(y_test, y_pred_nb) * 100, 2), "%")
print("\nClassification Report (Naive Bayes):\n", classification_report(y_test, y_pred_nb))
```

"""### 5.3 K-Nearest Neighbors (KNN)"""

```
knn_model = KNeighborsClassifier(n_neighbors=5)
knn_model.fit(X_train, y_train)
y_pred_knn = knn_model.predict(X_test)

print("Confusion Matrix (KNN):\n", confusion_matrix(y_test, y_pred_knn))
print("\nAccuracy (KNN):", round(accuracy_score(y_test, y_pred_knn) * 100, 2), "%")
print("\nClassification Report (KNN):\n", classification_report(y_test, y_pred_knn))
```

"""### 5.4 Decision Tree"""

```
dt_model = DecisionTreeClassifier(random_state=42)
dt_model.fit(X_train, y_train)
y_pred_dt = dt_model.predict(X_test)

print("Confusion Matrix (Decision Tree):\n", confusion_matrix(y_test, y_pred_dt))
print("\nAccuracy (Decision Tree):", round(accuracy_score(y_test, y_pred_dt) * 100, 2), "%")
print("\nClassification Report (Decision Tree):\n", classification_report(y_test, y_pred_dt))
```

"""### 5.5 Random Forest"""

```
rf_model = RandomForestClassifier(n_estimators=100, random_state=42)
rf_model.fit(X_train, y_train)
y_pred_rf = rf_model.predict(X_test)

print("Confusion Matrix (Random Forest):\n", confusion_matrix(y_test, y_pred_rf))
print("\nAccuracy (Random Forest):", round(accuracy_score(y_test, y_pred_rf) * 100, 2), "%")
print("\nClassification Report (Random Forest):\n", classification_report(y_test, y_pred_rf))
```

"""## 6. Model Comparison (Evaluation Summary)"""

```
results = {
'SVM': accuracy_score(y_test, y_pred_svm),
'Naive Bayes': accuracy_score(y_test, y_pred_nb),
'KNN': accuracy_score(y_test, y_pred_knn),
'Decision Tree': accuracy_score(y_test, y_pred_dt),
'Random Forest': accuracy_score(y_test, y_pred_rf),
}

comparison_df = pd.DataFrame(list(results.items()), columns=['Model', 'Accuracy'])
comparison_df['Accuracy (%)'] = (comparison_df['Accuracy'] * 100).round(2)
```

```
comparison_df = comparison_df.sort_values('Accuracy (%)', ascending=False)
comparison_df

plt.figure(figsize=(9, 5))
sns.barplot(data=comparison_df, x='Accuracy (%)', y='Model', hue='Model', palette='mako', legend=False)
plt.title("Model Accuracy Comparison")
plt.xlabel("Accuracy (%)")
plt.tight_layout()
plt.show()

"""## 7. Prediction

Best-performing model (highest accuracy) use karke test set ke pehle 10 samples ki prediction actual values ke saath compare kar rahe hain.
"""

best_model_name = comparison_df.iloc[0]['Model']
model_lookup = {
'SVM': svm_model,
'Naive Bayes': nb_model,
'KNN': knn_model,
'Decision Tree': dt_model,
'Random Forest': rf_model,
}
best_model = model_lookup[best_model_name]
print(f"Best Model: {best_model_name}")

y_pred_best = best_model.predict(X_test)

result_table = X_test.copy()
result_table['Actual_Match_Result'] = label_encoders['Match_Result'].inverse_transform(y_test)
result_table['Predicted_Match_Result'] = label_encoders['Match_Result'].inverse_transform(y_pred_best)

result_table[['Actual_Match_Result', 'Predicted_Match_Result']].head(10)

"""## 8. Visualizations (Matplotlib)"""

### Scatter Plot - Runs vs Strike Rate colored by Match Result ###
plt.figure(figsize=(8, 6))
sns.scatterplot(data=df, x='Runs', y='Strike_Rate', hue='Match_Result', palette='Set1', alpha=0.6)
plt.title("Scatter Plot: Runs vs Strike Rate (by Match Result)")
plt.show()

### Histograms of Key Numerical Attributes (Grid Layout) ###
import matplotlib.pyplot as plt
import seaborn as sns

hist_cols = ['Runs', 'Balls_Faced', 'Strike_Rate', 'Economy_Rate', 'Player_Impact_Score']

colors = ['red', 'green', 'blue', 'orange', 'purple']

fig, axes = plt.subplots(2, 3, figsize=(15, 8))
axes = axes.flatten()

for i, col in enumerate(hist_cols):
sns.histplot(
```

```
df[col],
bins=30,
kde=True,
ax=axes[i],
color=colors[i],
edgecolor='black',
alpha=0.8
)
axes[i].set_title(f"Histogram of {col}", fontsize=11)
axes[i].grid(True, linestyle='--', alpha=0.5)

axes[5].axis('off')

plt.tight_layout()
plt.show()
```

Box Plot - Match_Result distribution

```
plt.figure(figsize=(6, 4))
sns.boxplot(y=df['Match_Result'], color='skyblue')
plt.title("Box Plot of Match_Result (encoded)", fontsize=14)
plt.ylabel("Match_Result")
plt.tight_layout()
plt.show()
```

Box Plots without Outliers - Key Numeric Stats

```
numerical_columns = ['Runs', 'Balls_Faced', 'Strike_Rate', 'Player_Impact_Score', 'Target_Score']
fig, axes = plt.subplots(nrows=1, ncols=5, figsize=(18, 5))
for idx, col in enumerate(numerical_columns):
data = df[col].dropna()
axes[idx].boxplot(data, showfliers=False)
axes[idx].set_title(col)
plt.tight_layout()
plt.show()
```

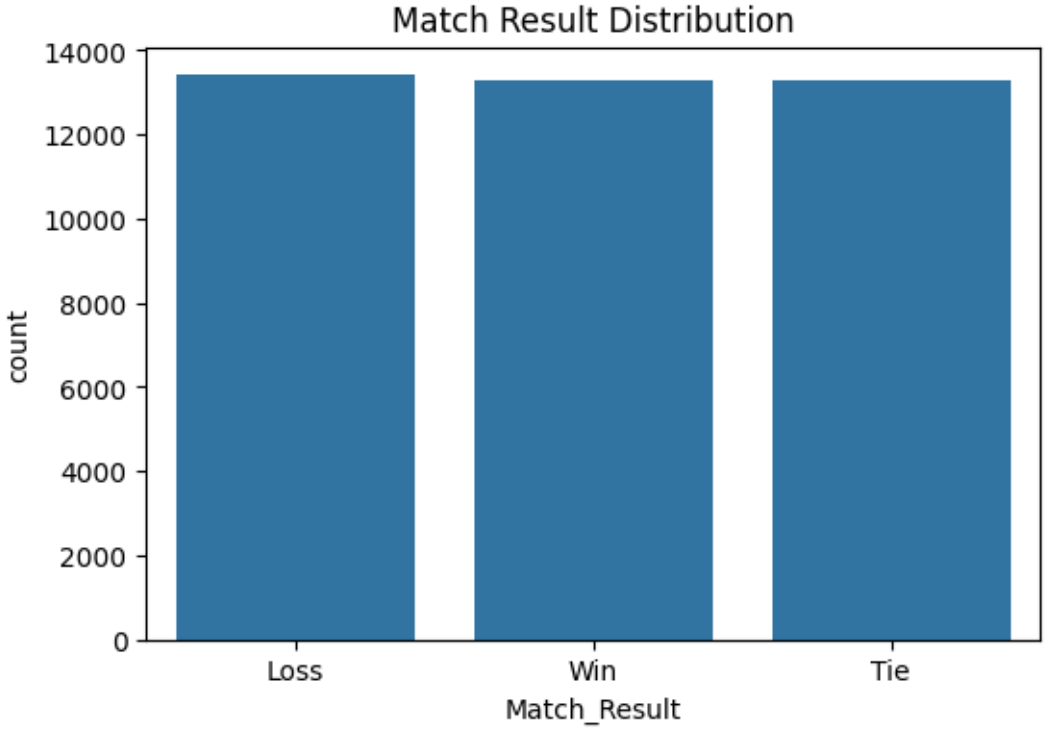
Confusion Matrix Heatmaps for all models

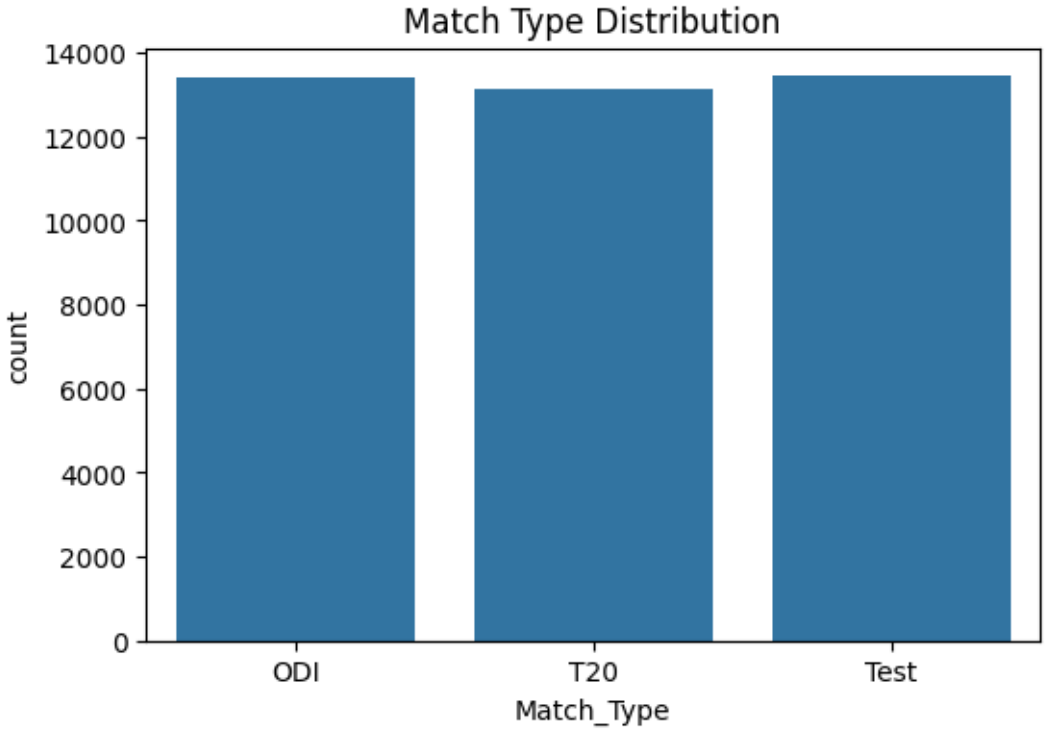
```
fig, axes = plt.subplots(2, 3, figsize=(16, 9))
axes = axes.flatten()

preds = {
'SVM': y_pred_svm,
'Naive Bayes': y_pred_nb,
'KNN': y_pred_knn,
'Decision Tree': y_pred_dt,
'Random Forest': y_pred_rf,
}

for i, (name, pred) in enumerate(preds.items()):
cm = confusion_matrix(y_test, pred)
sns.heatmap(cm, annot=True, fmt='d', cmap='Blues', ax=axes[i], cbar=False)
axes[i].set_title(name)
axes[i].set_xlabel("Predicted")
axes[i].set_ylabel("Actual")

axes[5].axis('off')
plt.tight_layout()
plt.show()
```

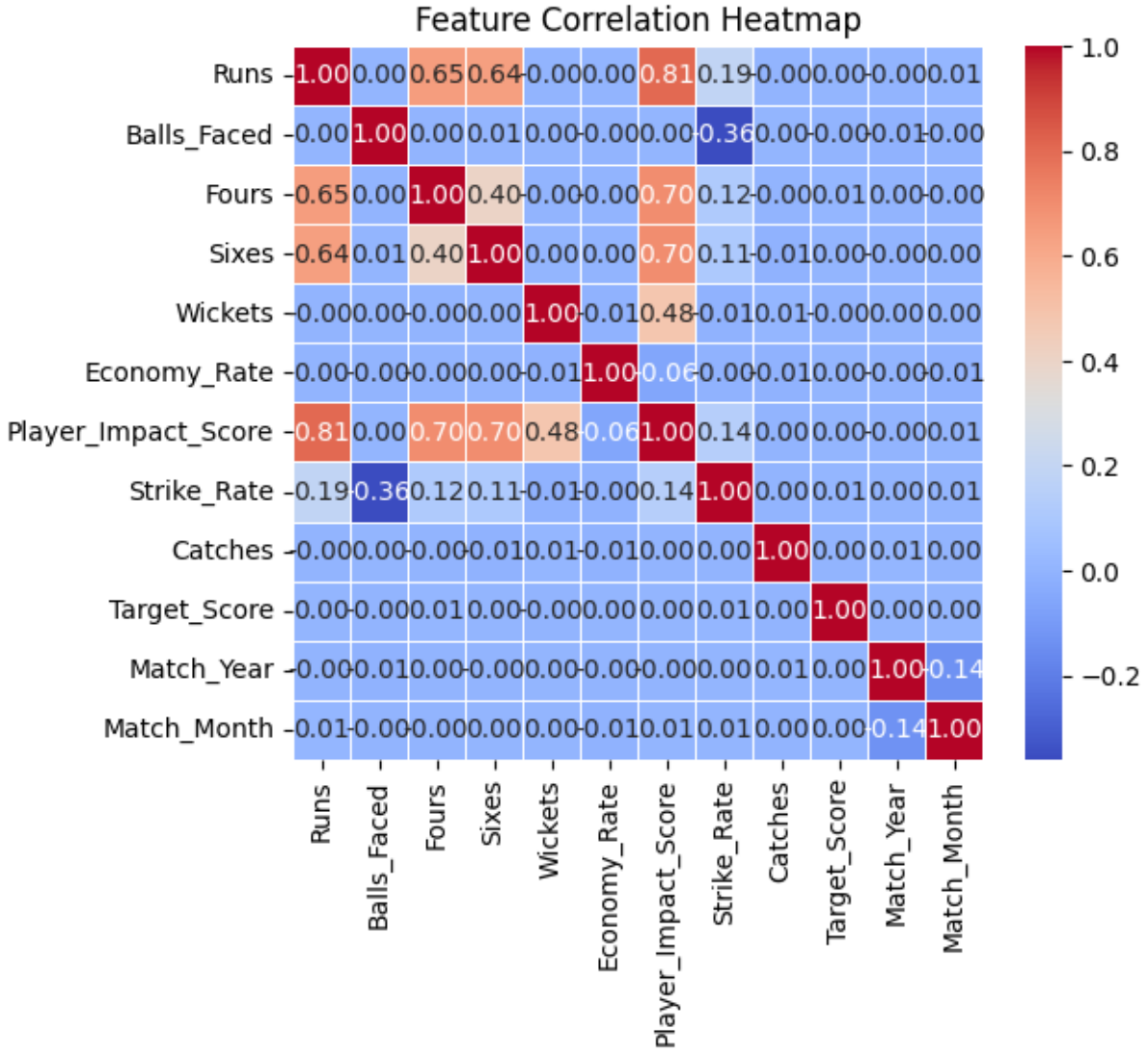
- **Output :**

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 40000 entries, 0 to 39999
Data columns (total 16 columns):
 #   Column              Non-Null Count  Dtype
---  ------              --------------  -----
 0   Date                40000 non-null  object
 1   Player_Name         40000 non-null  object
 2   Team                40000 non-null  object
 3   Opponent            40000 non-null  object
 4   Runs                40000 non-null  int64
 5   Balls_Faced         40000 non-null  int64
 6   Fours               40000 non-null  int64
 7   Sixes               40000 non-null  int64
 8   Wickets             40000 non-null  int64
 9   Economy_Rate        40000 non-null  float64
 10  Match_Result        40000 non-null  object
 11  Match_Type          40000 non-null  object
 12  Player_Impact_Score  40000 non-null  float64
 13  Strike_Rate         40000 non-null  float64
 14  Catches             40000 non-null  int64
 15  Target_Score        40000 non-null  int64
dtypes: float64(3), int64(7), object(6)
memory usage: 4.9+ MB

Shape: (40000, 16)

Missing values per column:
 Date                0
Player_Name          0
Team                 0
Opponent             0
Runs                 0
Balls_Faced          0
Fours                0
Sixes                0
Wickets              0
Economy_Rate         0
Match_Result         0
Match_Type           0
Player_Impact_Score  0
Strike_Rate          0
Catches              0
Target_Score         0
dtype: int64

Total missing values: 0

Duplicate rows: 0

Remaining missing values: 0
```

```
Match_Result
Loss    13415
Tie     13304
Win     13281
Name: count, dtype: int64
```







Match_Result classes: {0: 'Loss', 1: 'Tie', 2: 'Win'}
Training set: (28000, 16)
Testing set: (12000, 16)
Confusion Matrix (SVM):
[[1634 1066 1299]
[1650 1119 1309]
[1608 1016 1299]]

Accuracy (SVM): 33.77 %

Classification Report (SVM):

	precision	recall	f1-score	support
0	0.33	0.41	0.37	3999
1	0.35	0.27	0.31	4078
2	0.33	0.33	0.33	3923
accuracy			0.34	12000
macro avg	0.34	0.34	0.34	12000
weighted avg	0.34	0.34	0.34	12000

Confusion Matrix (Naive Bayes):
[[502 2260 1237]

```
 [ 488 2353 1237]
 [ 501 2256 1166]]
```

Accuracy (Naive Bayes): 33.51 %

Classification Report (Naive Bayes):

	precision	recall	f1-score	support
0	0.34	0.13	0.18	3999
1	0.34	0.58	0.43	4078
2	0.32	0.30	0.31	3923
accuracy			0.34	12000
macro avg	0.33	0.33	0.31	12000
weighted avg	0.33	0.34	0.31	12000

Confusion Matrix (KNN):

```
 [[1828 1339  832]
 [1833 1358  887]
 [1816 1303  804]]
```

Accuracy (KNN): 33.25 %

Classification Report (KNN):

	precision	recall	f1-score	support
0	0.33	0.46	0.39	3999
1	0.34	0.33	0.34	4078
2	0.32	0.20	0.25	3923
accuracy			0.33	12000
macro avg	0.33	0.33	0.32	12000
weighted avg	0.33	0.33	0.32	12000

Confusion Matrix (Decision Tree):

```
 [[1363 1308 1328]
 [1357 1374 1347]
 [1294 1319 1310]]
```

Accuracy (Decision Tree): 33.73 %

Classification Report (Decision Tree):

	precision	recall	f1-score	support
0	0.34	0.34	0.34	3999
1	0.34	0.34	0.34	4078
2	0.33	0.33	0.33	3923
accuracy			0.34	12000
macro avg	0.34	0.34	0.34	12000
weighted avg	0.34	0.34	0.34	12000

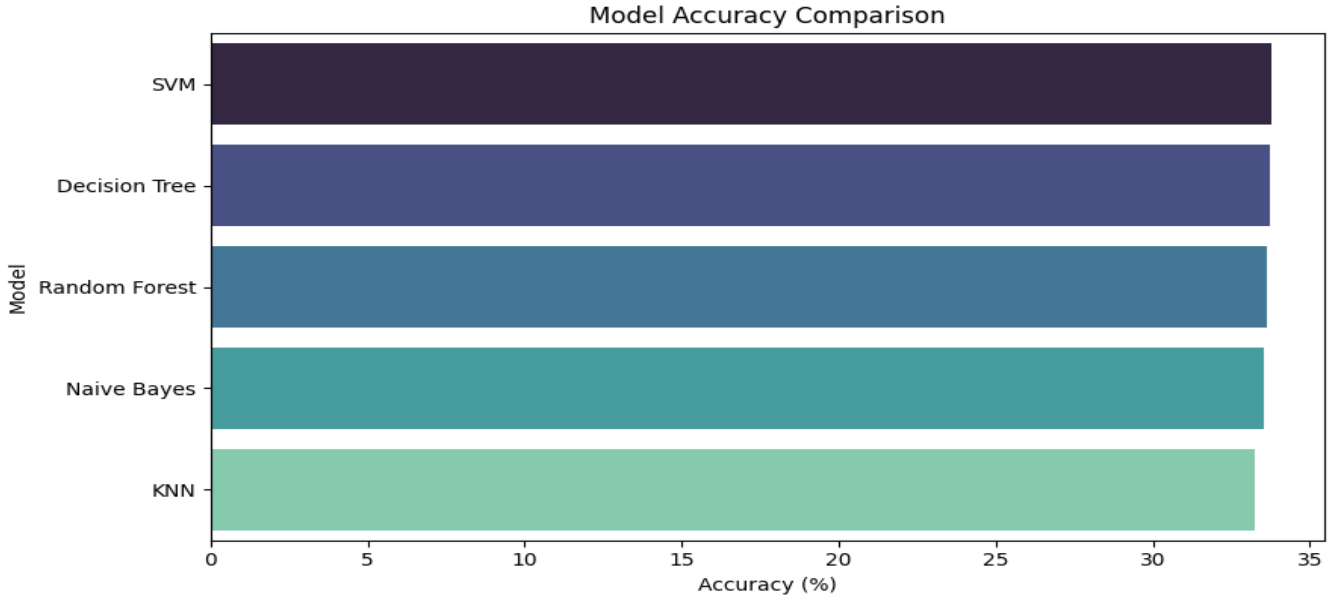
Confusion Matrix (Random Forest):

```
 [[1482 1241 1276]
 [1517 1271 1290]
 [1459 1179 1285]]
```

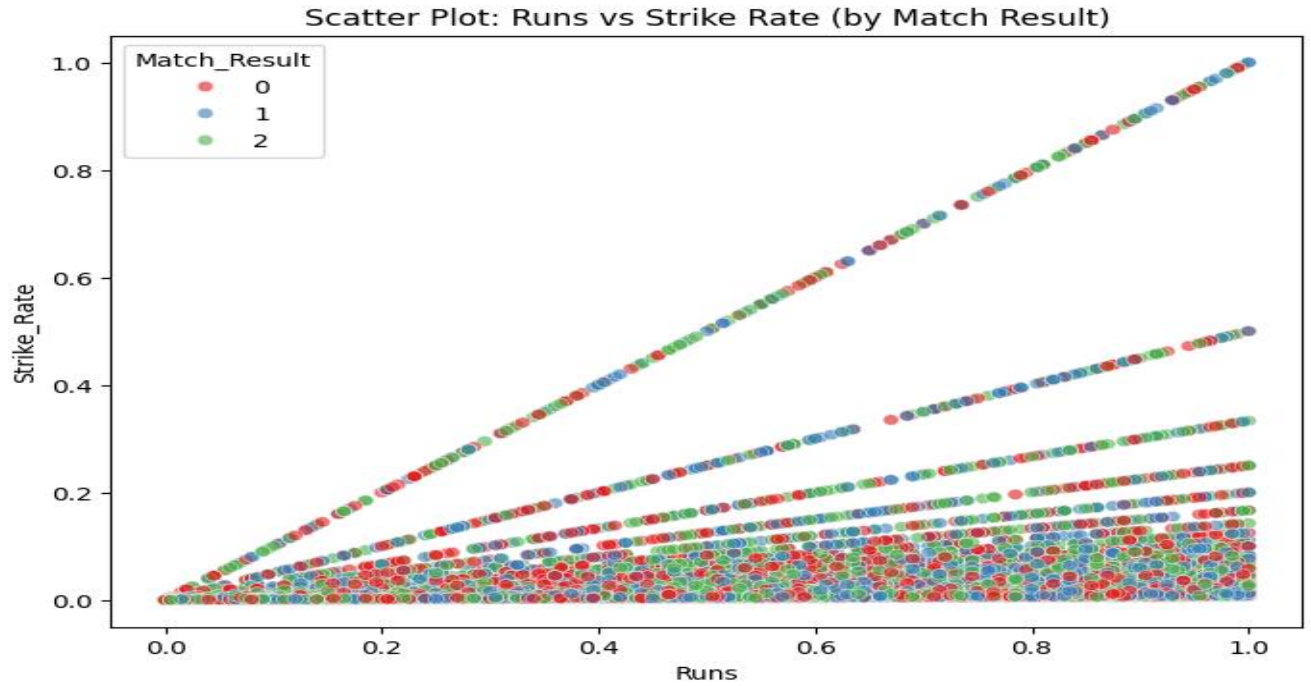
Accuracy (Random Forest): 33.65 %

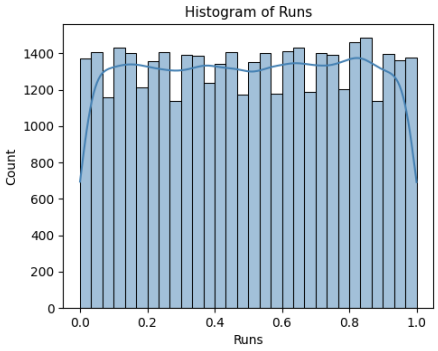
Classification Report (Random Forest):

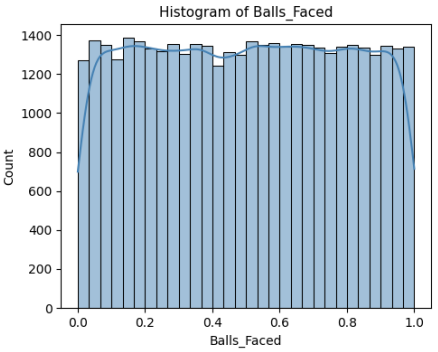
	precision	recall	f1-score	support
0	0.33	0.37	0.35	3999
1	0.34	0.31	0.33	4078
2	0.33	0.33	0.33	3923
accuracy			0.34	12000
macro avg	0.34	0.34	0.34	12000
weighted avg	0.34	0.34	0.34	12000

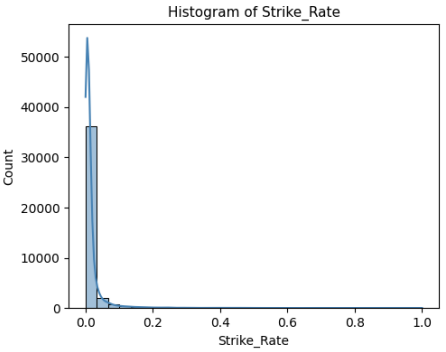


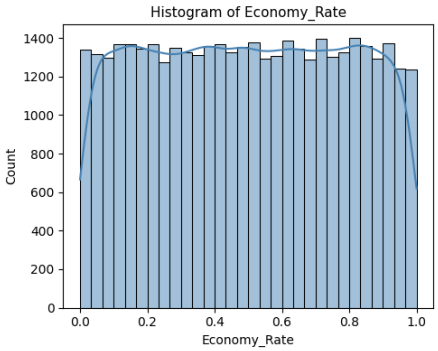
Best Model: SVM

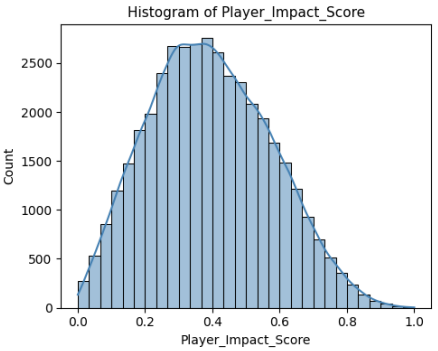


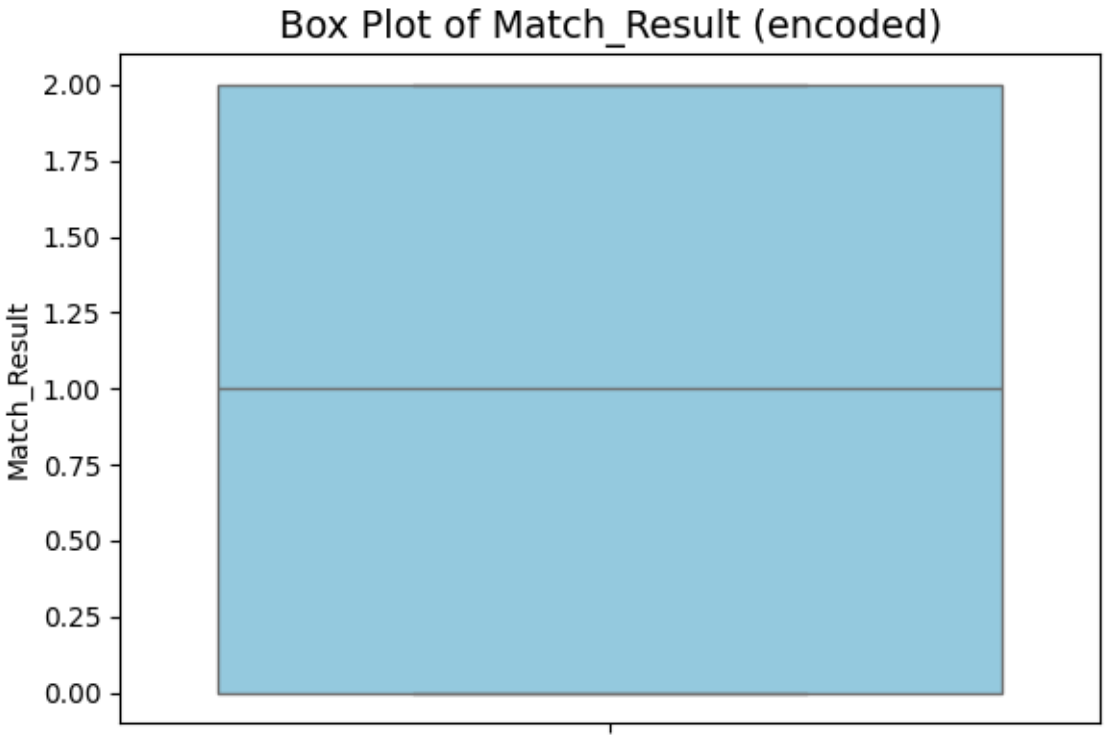

Histogram of Runs
Count
Runs

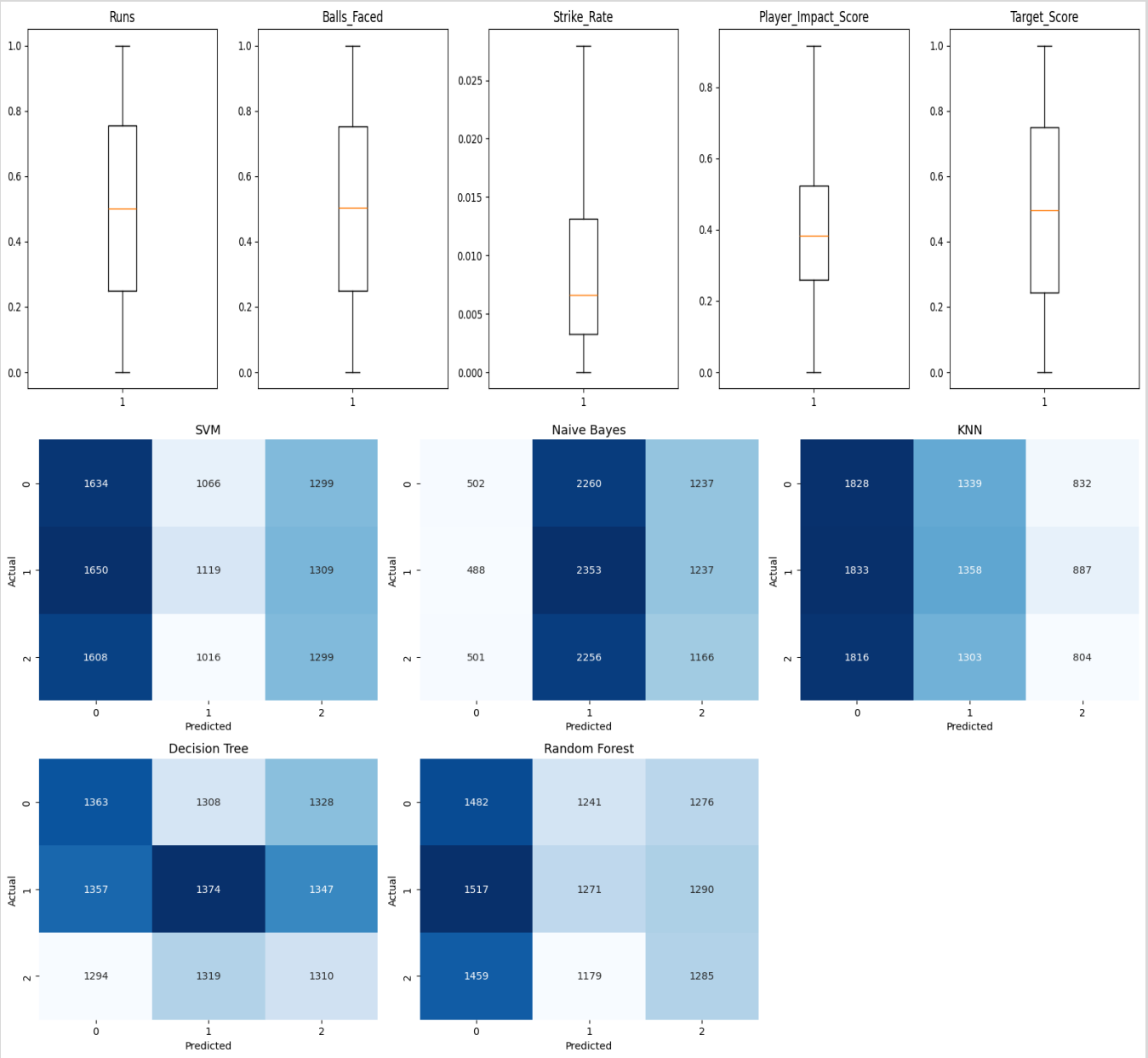

Histogram of Balls_Faced
Count
Balls_Faced


Histogram of Strike_Rate
Count
Strike_Rate


Histogram of Economy_Rate
Count
Economy_Rate


Histogram of Player_Impact_Score
Count
Player_Impact_Score


Box Plot of Match_Result (encoded)
Match_Result



4.2Test Report :

Each model was evaluated using Confusion Matrix, Accuracy Score, and Classification Report. The classification reports include Precision, Recall, F1-score, and Support.

- **Support Vector Machine (SVM):** Accuracy – **33.77%**
- **Naïve Bayes:** Accuracy – **33.51%**
- **K-Nearest Neighbors (KNN):** Accuracy – **33.25%**
- **Decision Tree:** Accuracy – **33.73%**
- **Random Forest:** Accuracy – **33.65%**

5. Proposed Enhancements:

The system can be improved by integrating real-time match data, advanced machine learning models, interactive dashboards, cloud deployment, and mobile applications. Additional features like weather analysis, player fitness tracking, and fantasy team recommendations can further enhance prediction accuracy and usability.

6. Conclusion :

The Cricket Player Performance Analysis system successfully analyzes player statistics using data science and machine learning techniques. It helps coaches, selectors, and analysts make better decisions by providing accurate performance insights through data analysis and visualization.

7. Bibliography :

1. Scikit-learn: Machine Learning in Python – https://scikit-learn.org/
2. Pandas Documentation – https://pandas.pydata.org/
3. NumPy Documentation – https://numpy.org/
4. Matplotlib Documentation – https://matplotlib.org/
5. Seaborn Documentation – https://seaborn.pydata.org/
6. Kaggle Cricket Dataset – https://www.kaggle.com/
7. ESPN Cricinfo Statistics – https://www.espncricinfo.com/
8. ICC Official Website – https://www.icc-cricket.com/